

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI, DUBAI CAMPUS
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI
SECOND SEMESTER 2022 – 2023
DEPARTMENT OF COMPUTER SCIENCE

Questions: 6
Pages: 3

Question Paper

COURSE : Machine Learning (BITS F464)

COMPONENT : Midsem Exam CLOSED BOOK

WEIGHTAGE : 30% (30 Marks)

DATE : 4-April-2023

DURATION : 90 Minutes (11:30 am- 1:00 pm)

Note:

1. Please write down any assumptions that you make.
2. Show the intermittant steps clearly for all answers.

Q. 1

Consider the following data related to possibility of heart disease based on presence of High BP, High Cholesterol, and High Sugar level.

5 M

Table 1: Data of possibility of heart disease

Patient ID	High BP	High Cholesterol	High Sugar level	Heart Disease
1	Yes	Yes	Yes	Yes
2	Yes	Yes	No	Yes
3	Yes	Yes	No	Yes
4	No	Yes	No	No
5	No	Yes	No	No
6	Yes	No	No	No
7	Yes	Yes	No	Yes
8	No	Yes	No	No
9	Yes	No	Yes	Yes
10	Yes	No	No	No
11	Yes	Yes	No	Yes

- i. What is the Gini Index of Heart Disease? Show all steps of Gini Index calculation. (2 M)
- ii. Which attribute will be chosen as the root of decision tree (based on Gini Index) by considering the data shown in Table 1. (Note: Heart disease as dependent variable)? Show all steps of decision of root node and also draw first level of decision tree. (4 M)

Q. 2

The University of Brilliance is working on classification model to classify the level of students into bright and average based on IQ test conducted for two subjects. The data consider for preparing model is as shown in Table 2.

5 M

- i. Which classification technique (other than logistic regression) will be best for the available data to better classify the level of students into bright and average? (1 M)
- ii. What will be the class label for level of student, if subject 1 and subject 2 marks are 20 and 21 respectively? Show all steps of decision of class label. (4 M)

Table 2: Data for prediction of Level of Students

Subject 1	Subject 1	Level of Students
20	22	Bright
18	19	Average
20	25	Bright
20	20	Average
17	14	Average
23	25	Bright
21	23	Bright

Q. 3

The Table 3 shows the examples of two numerical features $A1$ and $A2$ and binary response variable B . Build a linear discriminant analysis classifier and then classify the new instance ($A1=3.2$, $A2=5.3$). Show all steps of calculation.

8 M**Table 3: Data for Q.3**

A1	A2	B
2.9	6.6	1
2.5	7.7	1
3.5	5.6	1
3.1	5.4	1
2.5	4.4	0
2.1	6.2	0
3.2	3.5	0

Q. 4

i. Which of the following statements are true regarding Lasso regression? (2 M)

[Note: Question has multiple correct answers. Write what is reason for correct answers and also write why wrong option is not applicable for it.]

- Lasso's optimization problem can be stated as a quadratic program.
- The cost function minimized by Lasso has points where its gradient is not well-defined, and the solution (minimum) is often at such a point.
- Lasso often produces sparser results (more zero weights) than ridge regression.

d. A version of Lasso using a penalty term of $\lambda \|w\|_{\ell_{0.5}}$ (that is, the $\ell_{0.5}$ -norm) will be more inclined to produce sparse solutions than Lasso.

ii. Suppose you were to augment a decision tree by adding a Naive Bayes classifier at each leaf. That is, consider learning a decision tree of depth k , (where k is smaller than the number of variables n), where each leaf contains not a class, but a Naive Bayes classifier, which is based on the $n - k$ variables not appearing on the path to that leaf. Briefly outline a learning algorithm for this representation, assuming that k is a fixed parameter. (2 M)

iii. Consider a set of five training examples given as $((x_i, y_i), c_i)$ values, where x_i and y_i are the two attribute values (positive integers) and c_i is the binary class label: $\{((1, 1), -1), ((1, 7), +1), ((3, 3), +1), ((5, 4), -1), ((2, 5), -1)\}$. Classify a test example at coordinates (3, 6) using a k -NN classifier with $k = 3$ and

2 + 2 + 2 M

	Manhattan distance defined by $d((u, v), (p, q)) = u - p + v - q $. Your answer should be either +1 or -1. (Show all steps of calculation) (2 M)																																
Q.5	In a time of one year, the supermarket sold cauliflower at six different prices (x per kilograms). They calculated the average number of kilograms y sold per day at each of the six different prices. Following data is calculated from the available information, $\sum x = 200, \sum y = 436, \sum xy = 12515, \sum x^2 = 7250, \sum y^2 = 39234, n = 6$ Draw the regression line which best fit the given data. (Show all steps of calculation)	3 M																															
Q.6	Consider the following confusion matrix shown in Table 4 for four class classification, Table 4: Confusion matrix for four class classification <table><tr><th colspan="2" rowspan="2"></th><th colspan="4">Predicted Values</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><th rowspan="4">Actual Values</th><th>A</th><td>5</td><td>0</td><td>0</td><td>5</td></tr><tr><th>B</th><td>0</td><td>3</td><td>1</td><td>4</td></tr><tr><th>C</th><td>0</td><td>1</td><td>5</td><td>6</td></tr><tr><th>D</th><td>5</td><td>4</td><td>6</td><td>15</td></tr></table> Calculate precision, recall, F1-score, and accuracy. Show all steps of calculation.			Predicted Values				A	B	C	D	Actual Values	A	5	0	0	5	B	0	3	1	4	C	0	1	5	6	D	5	4	6	15	3 M
				Predicted Values																													
		A	B	C	D																												
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***** ALL THE BEST *****

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Questions: 8
Pages: 4

Question Paper

COURSE : Machine Learning (BITS F464)

COMPONENT : Endsem Exam CLOSED BOOK

WEIGHTAGE : 40% (40 Marks)

DATE : 30-May-2023

DURATION : 3 hrs (12:30 pm- 3:30 pm)

Note:

1. Please write down any assumptions that you make.
2. Show the intermittant steps clearly for all answers.

Q. 1

You have six training points (A, B, C, D, E, F, G) and five classifiers (H1, H2, H3, H4, H5) which make the following misclassifications:

5 M

Classifiers	Misclassified training points
H1	A, D, F, G
H2	D, E
H3	B, C, G
H4	A, B, F
H5	B, C, D, E

Perform two rounds (including initial round) of boosting with these classifiers and training data. In each round, pick the classifier with the lowest error rate. Break ties by picking the classifier that comes first in this list: H1, H2, H3, H4, H5. Show all steps of calculations and fill the following table,

	Round 1	Round 2
Weight _A		
Weight _B		
Weight _C		
Weight _D		
Weight _E		
Weight _F		
Weight _G		
Error rate of H1		
Error rate of H2		
Error rate of H3		
Error rate of H4		
Error rate of H5		
Weak classifier (h)		
Classifier error (ϵ)		
voting power (α)		

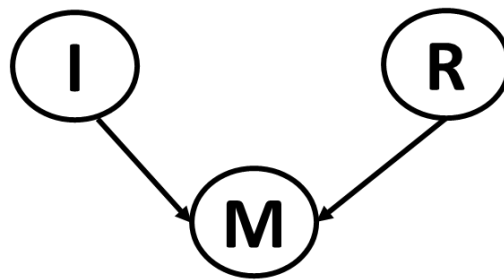
Q. 2	Find the cluster after two iterations of K-means clustering for the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9). Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2). Use distance measure as Manhattan Distance. Also find the initial mean, variance, and mixing coefficients for initial Gaussians to be used for EM for Gaussian mixture clustering.	5 M
Q. 3	Consider a following feature vectors, $x_1 = (2, 1)$, $x_2 = (3, 5)$, $x_3 = (4, 3)$, $x_4 = (5, 6)$, $x_5 = (6, 7)$, $x_6 = (7, 8)$. Use Principal Component Analysis (PCA) to reduce the dimension of feature vectors from 2 to 1. Show all steps of calculation of covariance matrix, eigen values, normalized eigen vectors, and sample points in one dimension with respect to Principal Component 1. Also draw the feature vectors in one dimension with respect to Principal Component 1.	6 M
Q. 4	Consider a following multilayer perceptron with n inputs, one hidden layer and output layer (Figure 1). The hidden and output layer uses linear pre-activation and sigmoid activation function respectively. Write step by step execution of forward and backward propagation. Assume cross entropy error function as loss function. The diagram shows a neural network with three layers. The input layer at the bottom consists of nodes labeled $x_1, x_2, x_3, \dots, x_M$ in blue circles. The hidden layer in the middle consists of nodes labeled $z_1, z_2, \dots, z_D$ in orange circles. The output layer at the top consists of a single node labeled y in a yellow circle. Arrows indicate fully connected forward propagation from the input layer to the hidden layer, and from the hidden layer to the output layer. Labels 'Input', 'Hidden Layer', and 'Output' are placed to the left of their respective layers. Figure 1: Multilayer Perceptron	5 M
Q. 5	Consider the following plotted sample data points, as shown in Figure 2. The scatter plot shows data points on a grid with x-axis from 0 to 9 and y-axis from 0 to 8. There are three black dots at coordinates (1, 2), (2, 3), and (3, 1). There are three red crosses at coordinates (6, 5), (7, 7), and (8, 6). Figure 2: Sample Data Points Find the support vectors and draw the decision boundary for linear SVM. (Show all steps of calculating discriminating hyperplane)	5 M

Q.6

During COVID outbreak you are worried about you might be infected and due to it and you are unable to write the endsem exam. For making this decision for either you are infected or not you decided to use Bayesian Net. You consider the following factors for making decision,

- You consider that you have immunity from the COVID (+i) due to recent vaccine you took.
- The vaccine you took is not effective enough so you might not be immune to COVID (-i).
- Your classmate has suffered from fever from last two days. Therefore, you are thinking you may suffer from COVID (+r) and not suffer from COVID (-r).

Let, consider that I and R are two random variables and M is another random variable takes the value $+m$ if you have COVID and $-m$ if you do not. By considering all the above description you wrote the following Bayesian Network to decide your chance of being not well and having COVID.



I	P(I)
+i	0.8

R	P(R)
+r	0.4

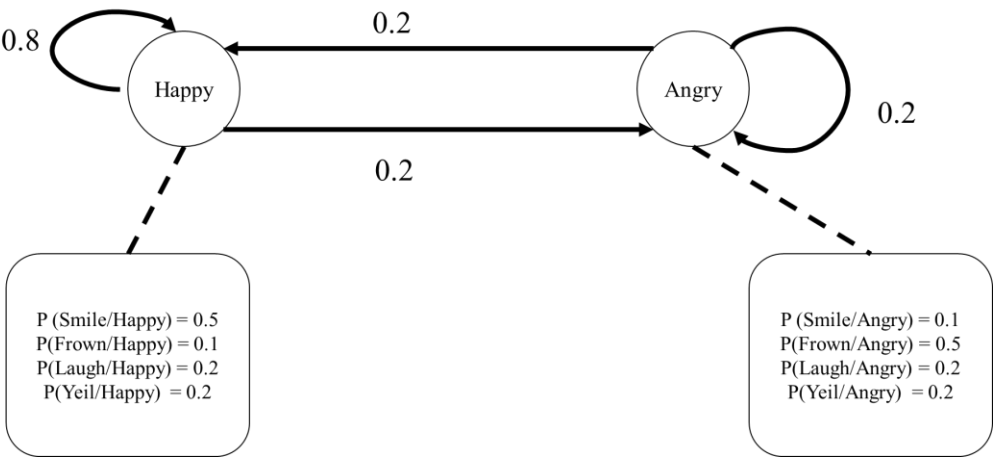
I	R	M	P(M/I,R)
+i	+r	+m	0
+i	+r	-m	1.0
+i	-r	+m	0
+i	-r	-m	1.0
-i	+r	+m	0.7
-i	+r	-m	0.3
-i	-r	+m	0.2
-i	-r	-m	0.8

Figure 3: Bayesian Network with probability table

Find the following probabilities based on Bayesian Network with probability table shown in Figure 3.

- $P(I = +i, R = +r, M = -m)$
- $P(I = +i, R = -r, M = -m)$
- $P(M = +m)$
- $P(R = +r / M = +m)$

6 M

Q.7	Children's are happy someday and angry on other days. We can only observe when they smile, frowns, laughs, or yells but not his actual emotional state. Let us start on day 1 in the happy state. There can be only one state transition per day. It can be either to happy state or angry state. The prior probabilities of Happy and Angry states are 0.7 and 0.3 respectively. The Hidden Markov Model of children state is as shown below,  Figure 4: Hidden Markov Model Assume that q_t is the state on day t and o_t is the observation on day t. Answer the following questions, - What is $P(o_2 = \text{frown})$? - What is $P(q_2 = \text{Happy} \mid o_2 = \text{frown})$? - What is $P(o_1 = \text{frown}, o_2 = \text{frown}, o_3 = \text{frown}, o_4 = \text{frown}, o_5 = \text{frown}, q_1 = \text{Happy}, q_2 = \text{Angry}, q_3 = \text{Angry}, q_4 = \text{Angry}, q_5 = \text{Angry})$?	5 M
Q.8	Answer the following questions, - Define Markov Decision Process. - What to do in the case of continuous value input for Naïve Bayes Classifier? - How regularization prevents overfitting?	3 M

***** ALL THE BEST *****